Biomedical Named Entity Recognition for Clinical Notes: A Comparative Study

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1 Introduction

In this project, Named Entity Recognition (NER) was applied to clinical notes in order to extract biomedical entities that will later be used to construct a clinical knowledge graph. Three different approaches were explored and compared: SciSpacy, BioBERT/PubMedBERT based models, and transformer-based chemical detection models.

1.1 SciSpacy Pipeline

The first approach used the SciSpacy model `en_ner_bc5cdr_md`. SciSpacy is a biomedical extension of the spaCy natural language processing library designed for scientific and medical text processing (Neumann *et al.*, 2019).

This model is trained on the BC5CDR biomedical dataset and is capable of detecting two important entity types: diseases and chemicals.

The SciSpacy pipeline was straightforward to implement. Each clinical note was processed using the spaCy pipeline, and the detected entities were extracted directly from the model output. The extracted entities were saved into a JSONL file that includes the entity text, label, and character positions in the original note.

During testing, SciSpacy performed reliably and was able to process long clinical notes without any input length limitations. The entity spans were generally clean and required minimal post-processing.

1.2 Transformer-Based NER Pipeline

To compare with SciSpacy, a transformer-based NER pipeline was implemented using models available through the HuggingFace Transformers library (Wolf *et al.*, 2020). Transformer models are based on the BERT architecture, which is widely used for natural language processing tasks (Devlin *et al.*, 2019).

Two models were used in the transformer pipeline:

- A PubMedBERT-based model for detecting disease entities
- A chemical detection model for identifying medications and other chemical entities

PubMedBERT is a biomedical language model trained entirely on biomedical literature from PubMed (Gu *et al.*, 2021). Similar biomedical transformer models such as BioBERT have shown strong performance in biomedical text mining tasks (Lee *et al.*, 2020).

Since these models were trained on different datasets, a single transformer model was not able to detect both diseases and chemicals reliably. Therefore, the outputs of the two models were combined to produce the final entity list.

The transformer models were applied to the same dataset of clinical notes, and the extracted entities were saved in the same JSONL format as the SciSpacy pipeline.

1.3 Implementation Challenges

Several practical challenges were encountered while implementing the transformer-based pipeline.

First, transformer models have a maximum input size of approximately 512 tokens due to the underlying BERT architecture (Devlin *et al.*, 2019). Some clinical notes were longer than this limit, which caused runtime errors. To address this issue, a preprocessing function was implemented to truncate very long notes before sending them to the model.

Second, transformer tokenization sometimes splits words into subword tokens. For example, a medication name such as "lisinopril" may be split into multiple pieces. A custom post-processing function was implemented to merge these tokens back into a single word.

Third, some transformer models produced partial entity spans or false positive predictions. For example, the phrase "type 2 diabetes" was sometimes partially detected as "type" only.

1.4 Experimental Environment

All experiments were conducted on a system equipped with an NVIDIA RTX 3070 Ti Mobile GPU, an Intel-based CPU, and 16GB of RAM. The experiments were implemented using Python with the HuggingFace Transformers library and SciSpacy models.

Although a GPU was available, the majority of experiments were executed on CPU due to compatibility and memory considerations when processing long clinical notes. Runtime performance therefore reflects a CPU-based execution environment.

1.5 Performance Comparison

During runtime evaluation, the transformer-based pipeline was significantly slower than the SciSpacy pipeline. Processing approximately 5000 clinical notes required more than eight hours when executed on CPU, with an average processing time of approximately 6 seconds per note.

In contrast, the SciSpacy pipeline processed the dataset much faster and required fewer additional preprocessing steps.

1.6 NER Model Comparison

Table 1: Comparison of NER models used in this project

Aspect	**SciSpacy**	**BioBERT / PubMedBERT**	**ClinicalBERT**
Model Type	Traditional biomedical NLP pipeline	Transformer (BERT-based)	Transformer (clinical-domain BERT)
Setup Complexity	Very simple pipeline setup	Moderate complexity	Moderate complexity
Entity Types	Disease and chemical entities	Disease and chemical using separate models	General clinical entities
Handling Long Notes	No input length limitation	Limited to $\sim$512 tokens	Limited to $\sim$512 tokens but pipeline handled it automatically
Tokenization Behaviour	Clean entity spans	Subword token splitting	Subword token splitting
Post-processing Required	Minimal	Token merging required	Token merging sometimes required
Runtime Performance	Fast	Slow on CPU	Slow on CPU
Engineering Complexity	Low	High	High
Stability in Clinical Notes	High	Moderate	Moderate
Overall Suitability	Efficient for large datasets	Flexible but computationally expensive	Useful for clinical language context

1.7 Summary of Findings

Overall, the results indicate that SciSpacy provides a more stable and efficient solution for large-scale biomedical entity extraction from clinical notes. The transformer-based pipeline offers flexibility and the potential for deeper contextual understanding, but it requires additional engineering steps and greater computational resources.

For this project, SciSpacy was therefore used as the primary NER pipeline, while the transformer-based approach was retained as an experimental comparison.

References

[1] Neumann, M., King, D., Beltagy, I. and Ammar, W., 2019. ScispaCy: Fast and Robust Models for Biomedical Natural Language Processing. Proceedings of the 18th BioNLP Workshop and Shared Task. Florence, Italy, pp.319–327. Available at: https://doi.org/10.18653/v1/W19-5034

[2] Devlin, J., Chang, M-W., Lee, K. and Toutanova, K., 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT), pp.4171–4186. Available at: https://doi.org/10.18653/v1/N19-1423

[3] Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C.H. and Kang, J., 2020. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. Bioinformatics, 36(4), pp.1234–1240. Available at: https://doi.org/10.1093/bioinformatics/btz682

[4] Gu, Y., Tinn, R., Cheng, H., Lucas, M., Usuyama, N., Liu, X., Naumann, T., Gao, J. and Poon, H., 2021. Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing. ACM Transactions on Computing for Healthcare, 3(1), pp.1–23. Available at: https://doi.org/10.1145/3458754

[5] Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., Cistac, P., Rault, T., Louf, R., Funtowicz, M. and Brew, J., 2020. Transformers: State-of-the-Art Natural Language Processing. Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, pp.38–45. Available at: https://doi.org/10.18653/v1/2020.emnlp-demos.6